

K633 — F445 Sector-Split Mechanism (PASS as understanding) + Harmonic-Ladder & Two-Axis Tau (LEADS, Cal #27)

Auditor: Keeper | Date: 2026-07-01 Wednesday | Verdict: split — see per-claim | Count: no motion (holds at 10)

Landing: Lyra F445 + Grace stratification/little-group scaffold + Elie toy checks, on the greenlit measurement-determinant lane (singlet-vs-colored root).

Claim 1 — Sector split forced by color charge (the F368 fiber-trace). PASS as structural understanding.

Mechanism: the measurement traces an N_c -dimensional fiber per mode; which fiber is the switch. Color-singlet (lepton) traces the geometric transverse fiber (curvature 2-forms $\rightarrow$ drags in π); color-triplet (down quark) traces the flat color fiber (bare index $\rightarrow$ integer, no π). Color is the switch that trades geometry-with- π for a plain integer count.

- Target-innocent: the switch is “colored vs singlet,” nothing tuned to a mass. Explains *why* $\mu\text{uon} = (24/\pi^2)^{C_2}$ (π -carrying) and down ladder = integer harmonics.
- Geometrically grounded: Grace’s scaffold locates it — singlet reads the bulk (Harish-Chandra deposit density $d(\nu)$, carries π^{N_c} and $5/2$); colored couples to the Shilov little group $SO(4)$ (integer S^4 harmonics). This is the F66/F86 bulk-boundary architecture applied to mass. Not a slogan.
- Verified: Elie 12/12 (toy 4524 disambiguation + checks).
- Ruling: this is the lane’s real deliverable — banked *as understanding*, not as count. Per team consensus (Keeper degeneracy + Cal #35), success here banks $\leq$ one mass ratio; the payoff is the sector-split mechanism. PASS.

Claim 2 — Cumulative- S^4 -harmonic reading of the down ladder. LEAD (strong), banks nothing yet.

Independent convergence: Grace (scaffold) and Lyra (F445) landed the same reading separately — the signal we trust most. Keeper-verified arithmetic: down-ladder counts are cumulative $SO(5)[\ell, 0]$ harmonic dims $\{1, 5, 14, 30\}$:
- $\text{gap}_{12} = N(\leq 2) = 1+5+14 = 20$ (vs physical ~ 20) - $\text{gap}_{23} = N(\leq 3) = 1+5+14+30 = 50$ (vs physical ~ 52 , 3.8%)
— beats $d(\nu)=64$ (+23%) and the dead artifact 45 (−13.5%). - Branching verified exact: $SO(5)[\ell, 0] = \bigoplus_{k \leq \ell} S^3\text{-harmonics, dims } (k+1)^2$ summing to the $SO(5)$ dim.

Why it stays a LEAD, not a bank (Grace + Lyra both flagged, correctly): - No a-priori reason a mass ratio should equal an absolute mode count. - No reason the degree cutoff increments per generation. - 20 is form-cheap (K631-S1). The whole reading rests on an unproven cutoff mechanism: does stratum geometry force the sequence $\{\leq 2, \leq 3\}$, and is the cutoff the generation-index k or the substrate primaries $\{\text{rank}, N_c\}$? Data cannot decide (Elie 4524); only Korányi-Wolf geometry can. $\rightarrow$ Grace’s lane (C3/C4). - If forced: down ladder banks at physical values (20, 50) with no running. If not: drop it.

Claim 3 — Two-axis (color $\times$ stratum-depth) tau-rescue. LEAD with Cal #27 flag — do not elevate.

Honest crack Lyra surfaced instead of smoothing: the tau breaks “all leptons are π ” — $m_\tau/m_e = 49.71$ is an *integer* lepton form. Grace’s two-axis rule resolves it: τ = color-singlet gen-3 sitting at the Shilov boundary $\rightarrow$ reads integer boundary modes even as a singlet; e, μ are singlets reading the bulk $\rightarrow \pi$ -forms.

- Keeper ruling: HOLD as lead, do NOT elevate. Grace self-fired Cal #27 correctly — two axes fitting four data points (e, μ, τ , down) is real fit-freedom. Peak-elegance is exactly where the bar fires hardest.
- Falsifier exists and must be checked first (Grace’s own prediction): light singlet neutrinos should be π /bulk; the rule predicts τ geometrically sits at Shilov while μ sits at bulk. Until the μ -bulk / τ -Shilov geometric assignment is *derived* (not fitted) and the neutrino prediction survives, this is a fit-shaped lead, not a mechanism.

- Good discipline: the crack was flagged, not smoothed; the rescue is grounded in the scaffold's bulk/boundary structure and is falsifiable. That earns "lead" status — not more.

Net

No count motion (holds at 10). Understanding-win (Claim 1) is real and audited PASS. Claims 2 and 3 are leads seated in the correct lane (Grace's Korányi-Wolf stratum geometry decides the cutoff-forcing question before any deep determinant build). Respects the ≤ 1 -bank ceiling.

— Keeper K633, Wednesday 2026-07-01. Sector-split PASS as understanding; harmonic-ladder strong LEAD pending cutoff-forcing; two-axis tau LEAD held under Cal #27 pending neutrino falsifier + derived stratum assignment.